CrewAnywhere2.0 — Master Architecture V4 (Short Version)

1. Document Information

Field	Value
Product	CrewAnywhere2.0
Document Type	Master Architecture (Short Version)
Platform Type	Marketplace + Workforce Operations Platform
Architecture Version	V4
Status	Canonical Approved Baseline
Primary Scope	Marketplace, Staffing, Operations, Payments
Canonical Sources	Master DB Architecture + Standardized FDR Standard

Based on canonical architecture references:

2. Platform Positioning

CrewAnywhere2.0 is positioned as:

- Marketplace + Workforce Operations Platform
- Event Workforce Operating Infrastructure
- Staffing + Operational Execution Ecosystem

The platform combines:

- Talent marketplace
- Hiring workflows
- Workforce assignment management
- Event-day operational execution
- Payment lifecycle infrastructure

3. Operational Hierarchy

Canonical operational hierarchy:

```
Event
└── Job
    └── Assignment
        └── Shift
```

Responsibility separation:

Layer	Responsibility
Event	Macro operational container

Job	Staffing lifecycle
Proposal	Hiring lifecycle
Assignment	Workforce engagement lifecycle
Shift	Daily operational execution lifecycle
Payment	Financial lifecycle

---

4. Architecture Principles

Core architecture principles:

- Proposal-centric hiring architecture
- Lean operational workflows
- Modular scalable architecture
- Event → Job → Assignment → Shift consistency
- Separation between hiring and operations
- Separation between assignment and shift execution
- Developer-ready backend structure
- Status-driven lifecycle enforcement

5. Platform Layers

Layer	Purpose
Account Layer	Authentication & account lifecycle
Marketplace Layer	Talent discovery & applications
Hiring Layer	Proposal lifecycle
Assignment Layer	Workforce engagement
Shift Layer	Operational execution
Payment Layer	Escrow, release, withdrawal
Compliance Layer	KYB, KYC, tax verification

6. Account Status Architecture

account_status ENUM

PENDING_VERIFICATION

ACTIVE

SUSPENDED

DELETED

Purpose:

- Authentication lifecycle
- Login access control
- Email verification lifecycle
- Suspension management

Applies to:

- Business Users
- Crew Users

---

7. Business Registration Architecture

Business registration flow:

Select User Type

→ Choose Signup Method

→ Validation

→ Account Creation

→ Database Creation

→ Email Verification

→ Dashboard Routing

Supported signup methods:

- Google
- LinkedIn
- Email

Registration includes:

- Existing account validation
- OAuth prefilled data
- Terms acceptance
- Verification handling

---

8. Business Readiness Architecture

business_ready Logic

Business Ready requires:

- Company Profile Completed
- Billing & Payment Setup Completed
- Tax Setup Completed
- KYB Submitted

verified_business Logic

Verified Business requires:

KYB Status = APPROVED

Purpose:

- Future compliance & trust layer
- Reserved for advanced verification systems

---

9. Event & Job Creation Architecture

Canonical architecture:

- Event ID = Parent Entity
- Job ID = Child Entity

Supports:

- Multi-role events
- Independent staffing fulfillment
- Multiple staffing requirements

Creation paths:

- Manual creation
- AI-assisted role creation

AI scope only applies to:

- Skills suggestions
- Budget suggestions
- Role requirement suggestions

Validation rule:

job_count >= 1

Required before publish flow continues.

---

10. Event Status Architecture

Event Status ENUM

Draft

Open

Closed

Definitions:

Status	Meaning
Draft	Event being prepared
Oper	Operationally active
Closed	Lifecycle finalized

---

11. Job Status Architecture

Job Status ENUM

Draft

Open

Reviewing

Filled

Active

Completed

Closed

Expired

Purpose:

- Staffing lifecycle management
- Assignment dependency enforcement
- Dashboard organization

---

12. Event ↔ Job Dependency

Event Status	Allowed Job Status
Draft	Draft
Open	Open, Reviewing, Filled, Active
Closed	Completed, Closed, Expired

13. Crew Registration Architecture

Crew registration flow:

Select User Type

→ Signup Method

→ Validation

→ Account Creation

→ Email Verification

→ Start Profile Creation

Supported methods:

- Google
- LinkedIn
- Email

Validation includes:

- Required fields
- Password strength
- Duplicate email validation
- Email format validation

14. Crew Profile Creation Architecture

Profile sources:

- LinkedIn Import
- Manual Form
- CV Upload

Profile sections:

A. Basic Info

- Name
- DOB
- Gender
- Profile Image
- Introduction

B. Job Preferences & Skills

- Categories
- Skill Tags

C. Professional Details

- Experience
- Certifications
- Rates
- Availability
- Languages

---

15. Marketplace Readiness Architecture

marketplace_ready Logic

Marketplace Ready requires:

- Email verified
- Required profile fields completed
- Profile published
- Minimum profile score achieved

Enables:

- Marketplace visibility
- Job applications
- Proposal submissions

16. Identity Verification Architecture

Supported IDs:

- Passport
- National ID
- Driver License
- Government IDs

Verification methods:

- Selfie verification
- Video verification
- Phone verification
- Proof of address

Verification lifecycle:

SUBMITTED

APPROVED

ADDITIONAL_INFO_REQUESTED

REJECTED

17. Marketplace & Job Discovery Architecture

Marketplace supports:

- Search-based discovery
- Filtering systems
- Best match sorting
- Recent jobs sorting

Separation logic:

- Saved Jobs separated from Applications
- Applications separated from Current Jobs

Current job tracking:

Applied

In Review

Hired

Completed

---

18. Proposal Lifecycle Architecture

Proposal Status ENUM

Applied

Offer Sent

Offer Accepted

Declined

Withdrawn

Hired

Architecture rule:

- Proposal lifecycle ends at Hired
- Assignment becomes authoritative operational object after hire

Proposal remains:

- Hiring lifecycle object
- Not operational execution object

---

19. Proposal Negotiation Architecture

Canonical negotiation flow:

Business Modifies Terms

→ Crew Reviews Terms

→ Accepted?

If accepted:

Finalize & Hire Crew

→ Payment Triggered

If rejected:

Continue Negotiation

→ Decline OR Withdraw

Important decisions:

- Dashboard = Job-centric
- Job Detail = Proposal-centric
- Payment only starts after finalized hire

---

20. Assignment Architecture

Assignment becomes active after:

- Proposal Status = HIRED
- Payment funded
- Workforce engagement confirmed

Assignment responsibilities:

- Workforce engagement
- Shift orchestration
- Operational scheduling

Canonical relationship:

jobs.job_id

↓

assignments.job_id

---

21. Assignment Status Architecture

Assignment Status ENUM

Scheduled

Active

Completed

Cancelled

Trigger logic:

Status	Trigger
Scheduled	Business hires crew
Active	Shift checked-in/in progress
Comp	All shifts completed

Completed	
Cancelled	Event/business cancellation

---

22. Job ↔ Assignment Dependency

Job Status	Assignment Behavior
Draft	No assignments
Open	No assignments
Reviewing	No assignments
Filled	Assignment scheduled

Active	Assignment active
Completed	Assignment completed
Closed	Assignment cancelled
Expired	None or cancelled

---

23. Shift Architecture

Canonical relationship:

assignments.assignment_id

↓

shifts.assignment_id

Shift records are generated from:

Assignment Creation

Purpose:

- Attendance tracking
- Operational execution
- Check-in/check-out workflows

---

24. Shift Status Architecture

Shift Status ENUM

Scheduled

Checked-In

In Progress

Completed

No-Show

Cancelled

Trigger examples:

Status	Trigger
Checked-In	QR verification
In Progress	Successful check-in
Completed	Check-out verification

No-Show	Attendance expired

---

25. Assignment ↔ Shift Dependency Architecture

Assignment Status	Allowed Shift Status
Scheduled	Scheduled
Active	Checked-In, In Progress
Completed	Completed
Cancelled	Cancelled, No-Show

---

26. Event-Day Operational Execution Architecture

Operational flow:

Shift Scheduled

→ Crew Check-In

→ QR Verification

→ In Progress

→ Check-Out

→ Supervisor Confirmation

→ Completed

Validation methods:

- QR verification
- Supervisor approval
- GPS validation
- Manual override

Purpose:

- Attendance verification
- Workforce tracking
- Payment release trigger

---

27. Payment Architecture

Payment lifecycle:

Milestone Funded

→ Work Submitted

→ Client Approval

→ Security Hold

→ Funds Available

→ Withdrawal

Payment architecture includes:

- Escrow funding
- Milestone release
- Refund handling
- Withdrawal processing

Payment starts only after:

Proposal Finalized + Hire Confirmed

---

28. Finance / Tax / Withdrawal Architecture

Finance architecture includes:

Tax

- tax_identification
- tax_forms

KYC

- legal_name
- government_id
- DOB
- address

Withdrawal Methods

- Bank Account
- PayPal
- Wise

---

29. Dashboard Architecture

Dashboard architecture:

Level 1

Job-centric dashboard

Level 2

Proposal-centric Job Detail

Level 3

Candidate Review + Chat + Hiring

Dashboard focuses on:

- Staffing operations
- Proposal counts
- Hiring progress
- Assignment tracking

---

30. AI-Assisted Staffing Architecture

AI only assists:

- Role/job creation
- Skills suggestions
- Budget estimation
- Requirement suggestions

Architecture rule:

> User remains authoritative editor

AI does not control:

- Final event data
- Final hiring decisions
- Operational execution

---

31. Documentation Standard Architecture

Official documentation standard:

Combined Lean FDR + Developer-Ready Technical FDR

Mandatory characteristics:

- Lean workflows
- Backend-ready structure
- ENUMs
- APIs
- Validation logic
- State transitions
- Security sections
- Acceptance criteria

---

32. Database Separation Architecture

CrewAnywhere2.0 is officially treated as:

Separate architecture baseline

Independent from:

- Legacy CrewAnywhere DB
- Legacy workflow structures
- Legacy status systems

---

33. Final Canonical Status Matrix

Layer	Statuses

Account	Pending Verification, Active, Suspended, Deleted
Event	Draft, Open, Closed
Job	Draft, Open, Reviewing, Filled, Active, Completed, Closed, Expired
Proposal	Applied, Offer Sent, Offer Accepted, Declined, Withdrawn, Hired
Assignment	Scheduled, Active, Completed, Cancelled
Shift	Scheduled, Checked-In, In Progress, Completed, No-Show, Cancelled

---

34. Final Operational Lifecycle

Canonical operational lifecycle:

Registration

→ Profile Creation

→ Marketplace Readiness

→ Event Creation

→ Job Creation

→ Marketplace Discovery

→ Proposal Submission

→ Negotiation

→ Hire

→ Assignment Creation

→ Shift Generation

→ Event-Day Execution

→ Payment Release

→ Withdrawal

---

35. Canonical Source of Truth

Official canonical architecture references:

1. CrewAnywhere2.0 Master Saved Memory & Database Architecture
2. CrewAnywhere2.0 Master Database Architecture Updated V3
3. CrewAnywhere2.0 Standardized Combined Lean FDR + Developer-Ready Technical FDR Standard